

4D COVERAGE COLLAPSE ANALYSIS REPORT

Date: February 05, 2026
Experiment: 4D Boolean Minimization Coverage Modeling

MODEL DESCRIPTION

This report analyzes the coverage collapse behavior of 4D cluster formation in Boolean function minimization across different minterm densities (0.3, 0.5, 0.7, 0.9).

The model predicts when adding more variables (n) stops improving coverage and begins producing redundant clusters.

MATHEMATICAL MODEL

$$\text{Coverage}(n) = (C_{19} \times r^k \times I_{\text{sat}}) / (\text{density} \times 2^n) \times 100$$

where:

- C_{19} : Number of 4D clusters at $n=19$
- r : Geometric growth ratio (cluster increase rate)
 - k : Counter ($k = n - 19$, starts at 1 for $n=20$)
- I_{sat} : Saturation information density (minterms/cluster)
 - n : Number of variables

PARAMETERS BY DENSITY

Density 0.3:

$C_{19} = 244$ clusters

$r = 2.1186$

$I_{\text{sat}} = 5.92$ minterms/cluster

Density 0.5:

$C_{19} = 388$ clusters

$r = 2.2748$

$I_{\text{sat}} = 9.05$ minterms/cluster

Density 0.7:

$C_{19} = 845$ clusters

$r = 2.2432$

$I_{\text{sat}} = 16.05$ minterms/cluster

Density 0.9:

$C_{19} = 4755$ clusters

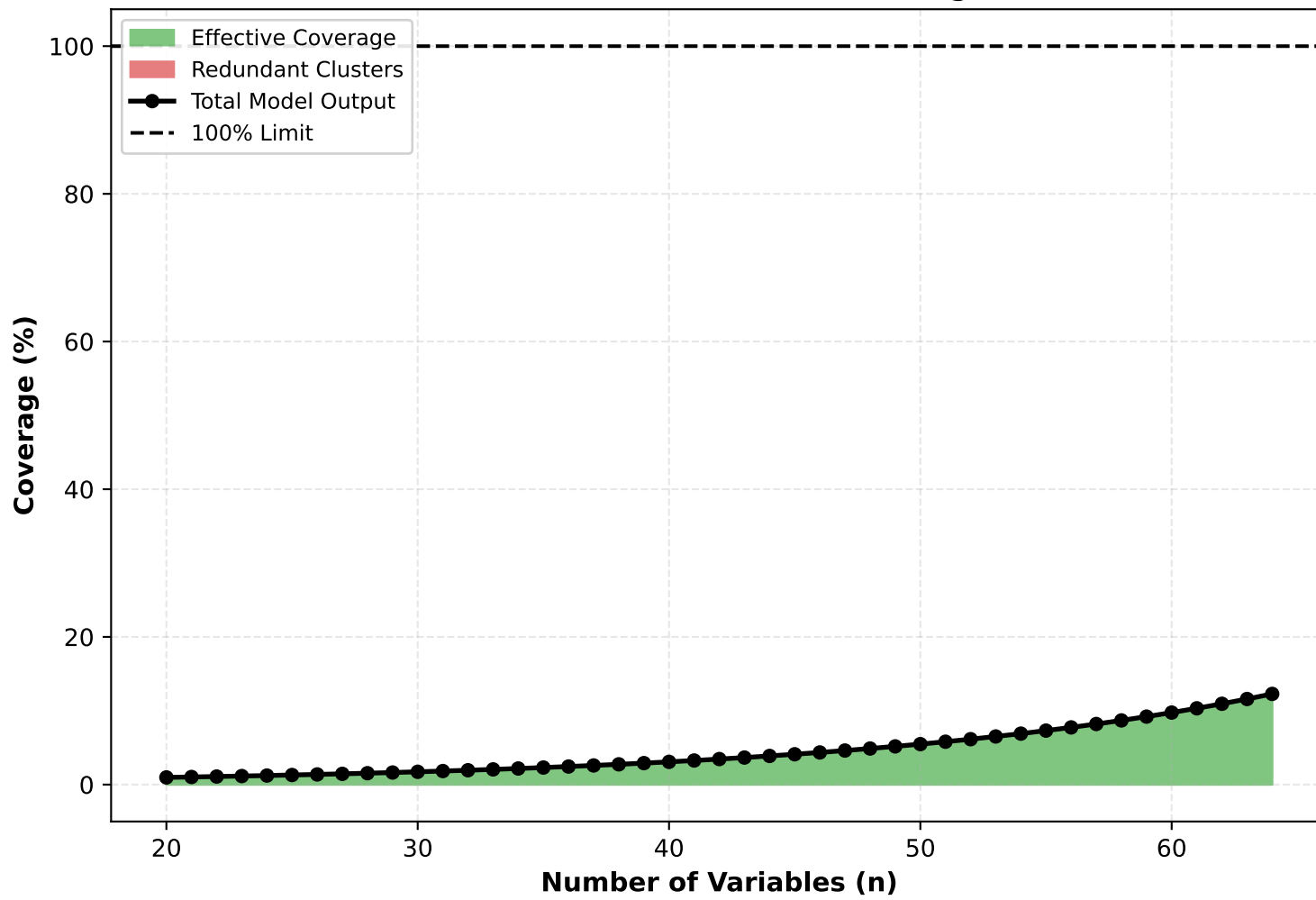
$r = 2.1834$

$I_{\text{sat}} = 33.60$ minterms/cluster

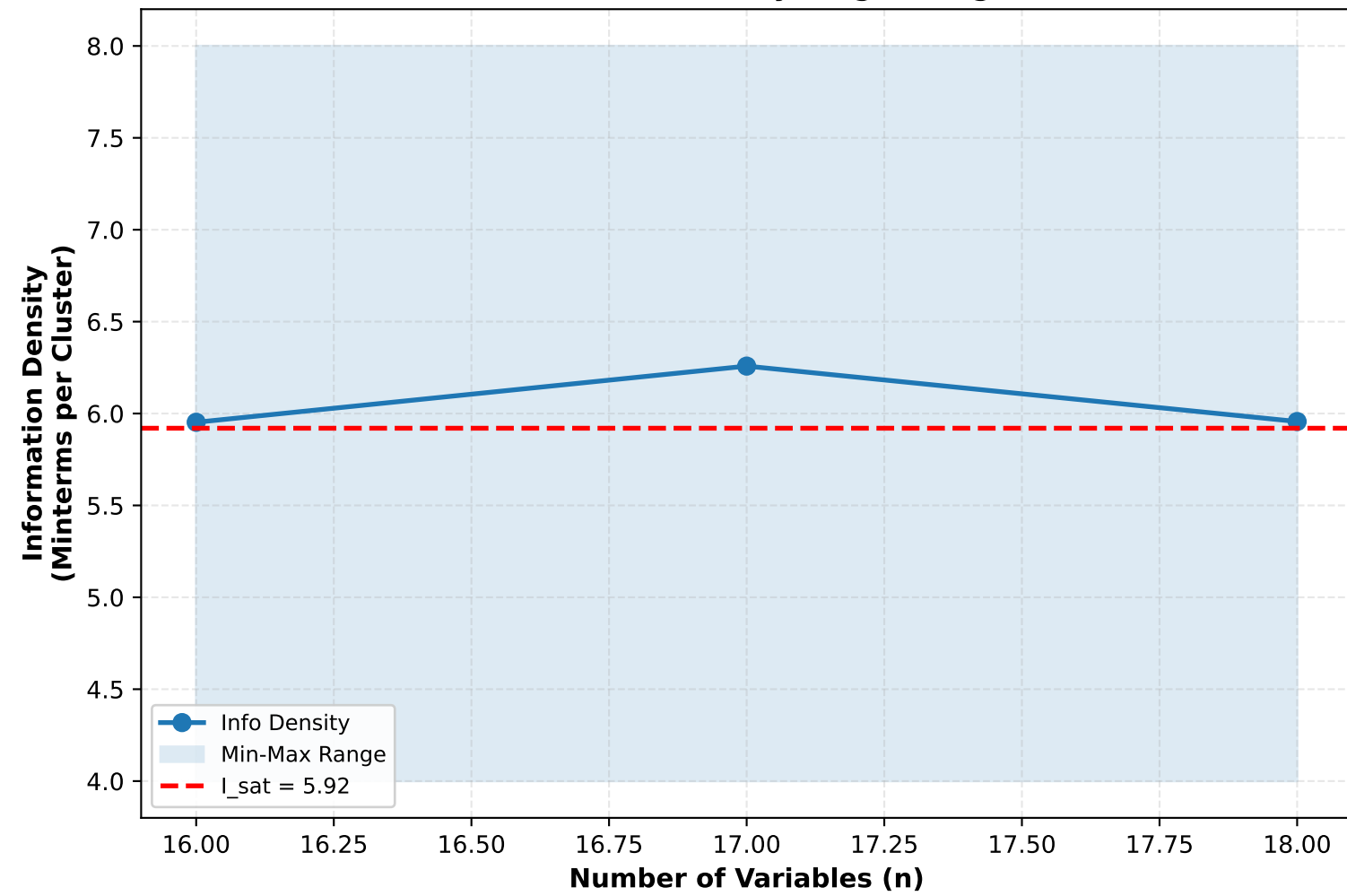
Analysis follows on subsequent pages, one per density.

4D Cluster Analysis - Density 0.3

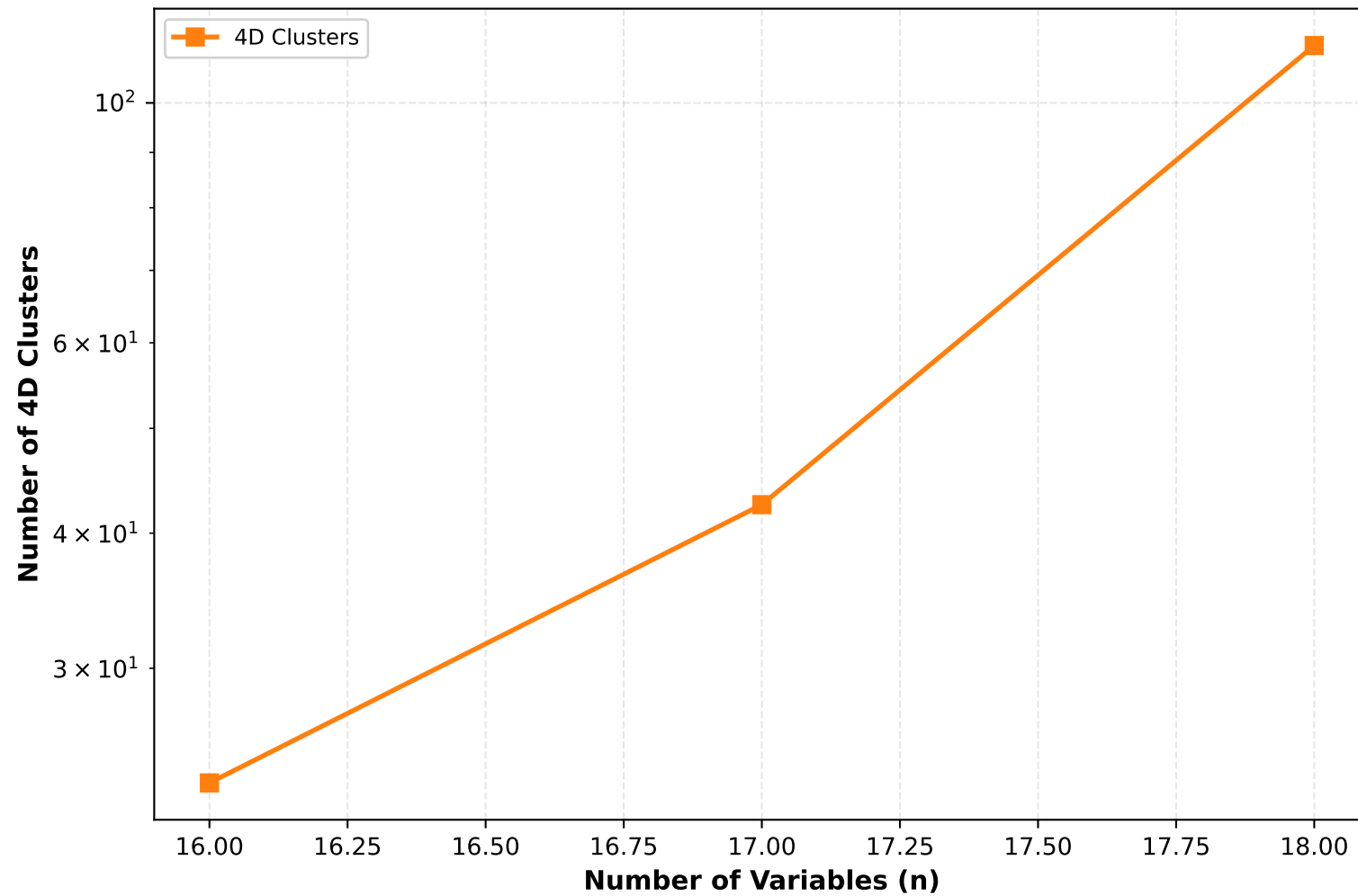
Effective vs Redundant Coverage



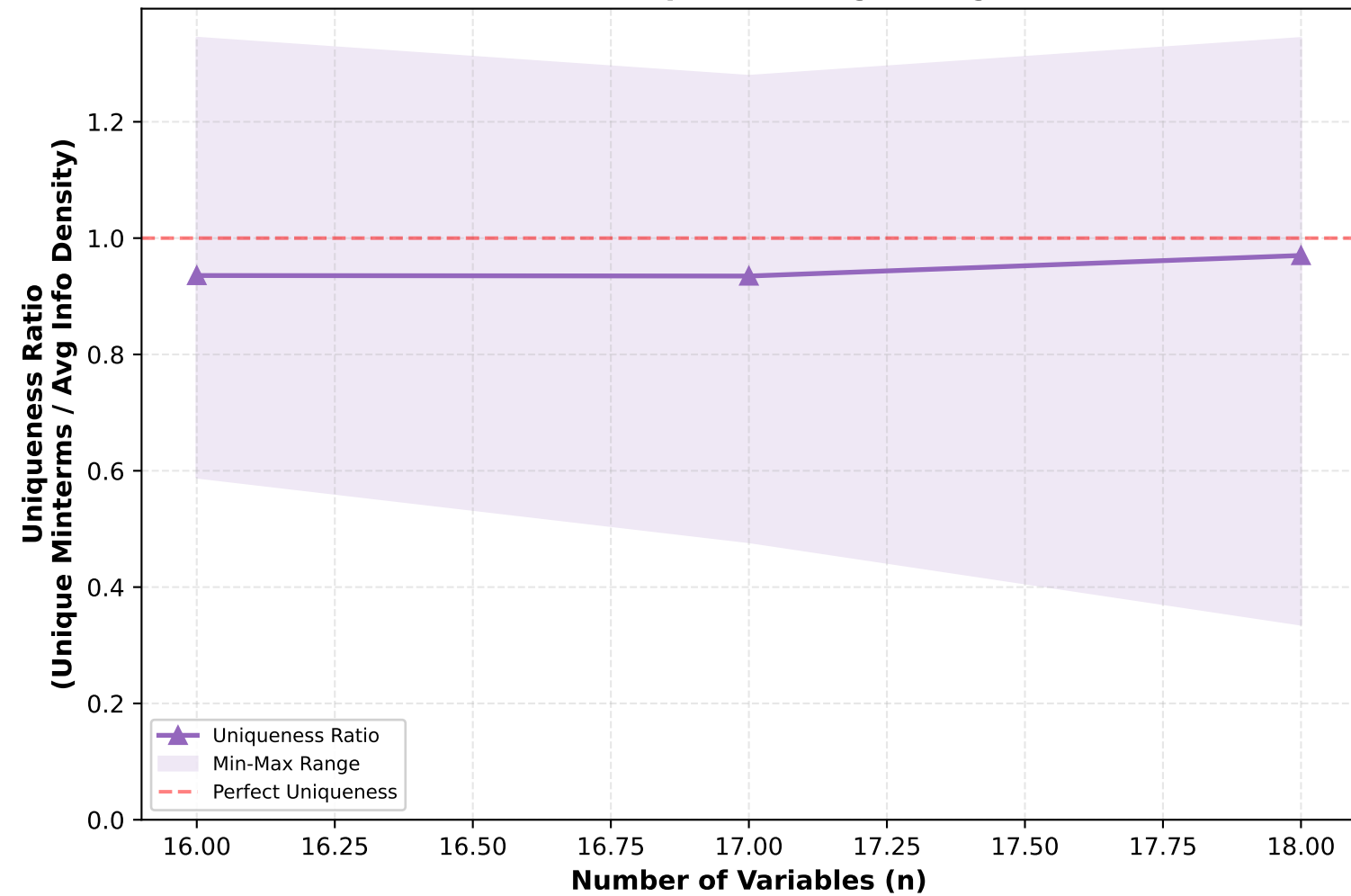
Information Density (avg $\pm$ range)



Cluster Growth

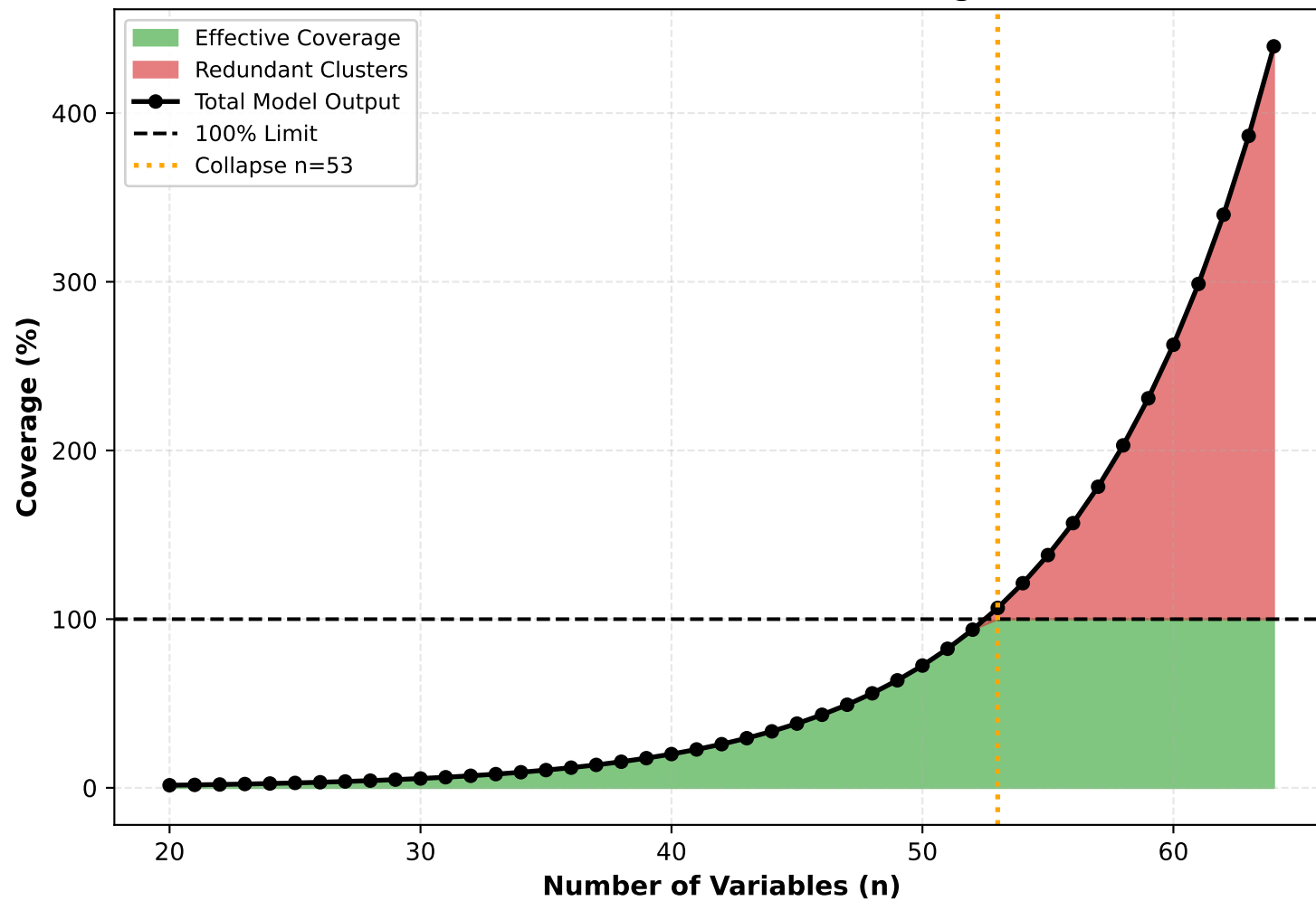


Cluster Uniqueness (avg $\pm$ range)

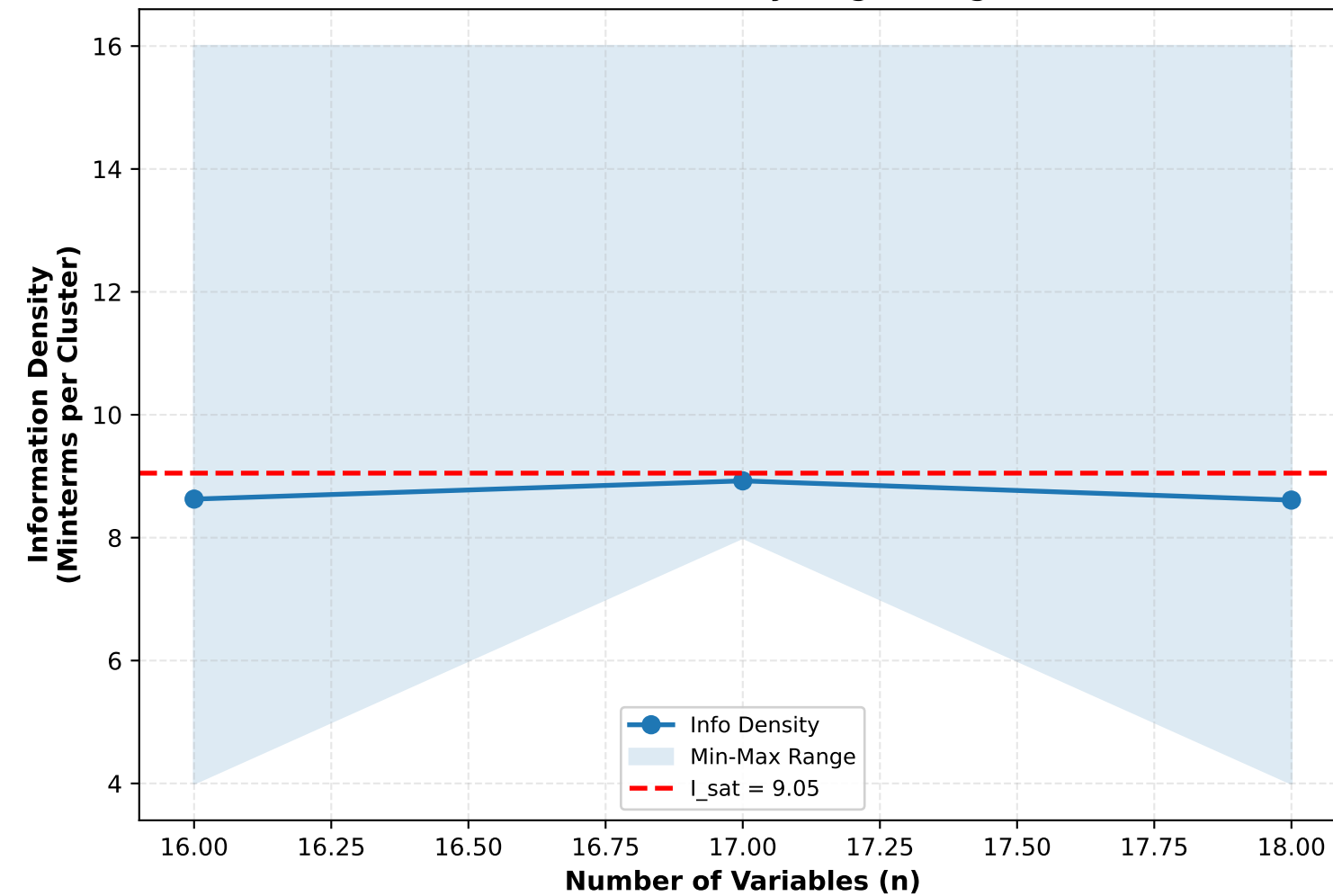


4D Cluster Analysis - Density 0.5

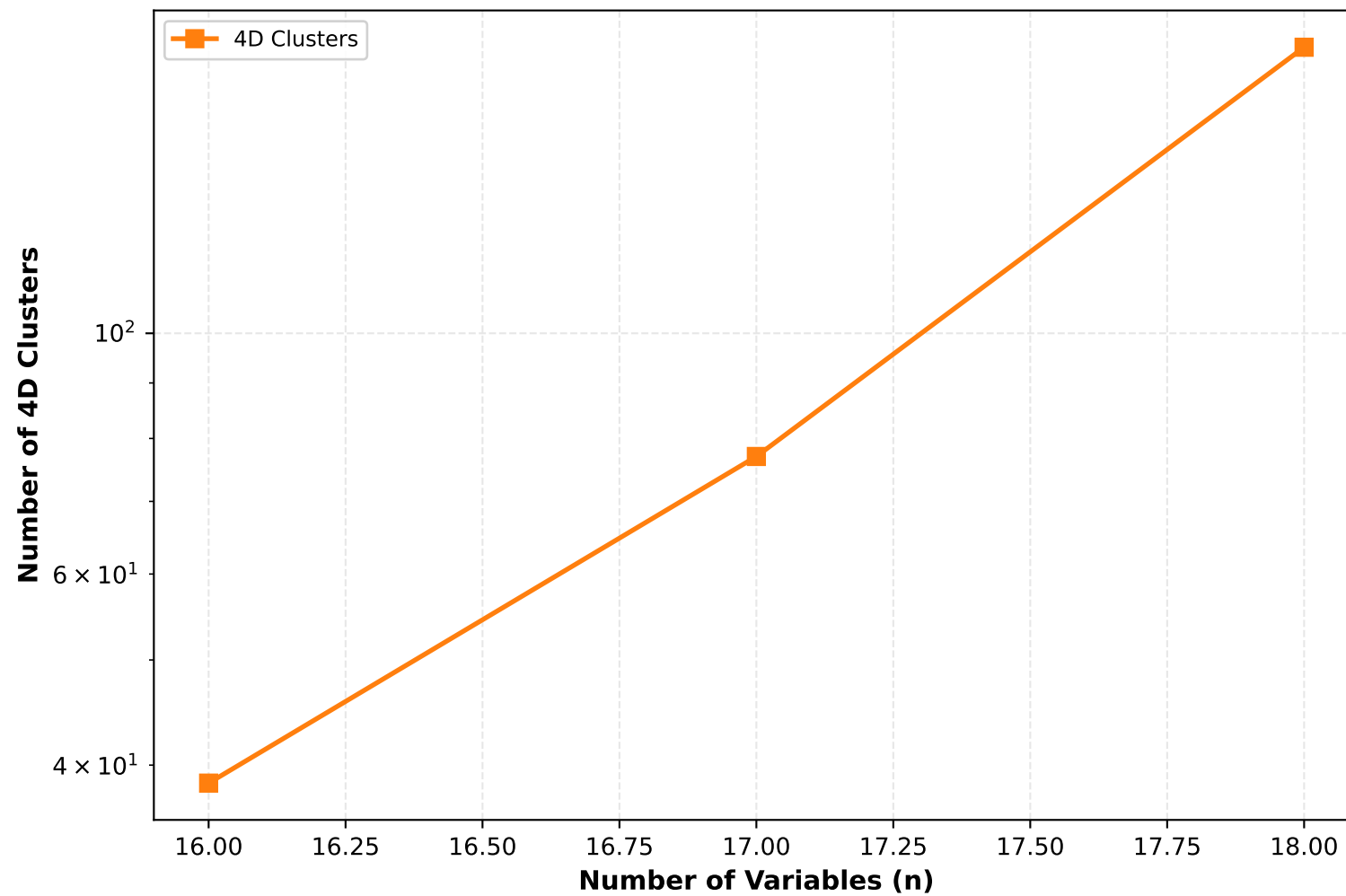
Effective vs Redundant Coverage



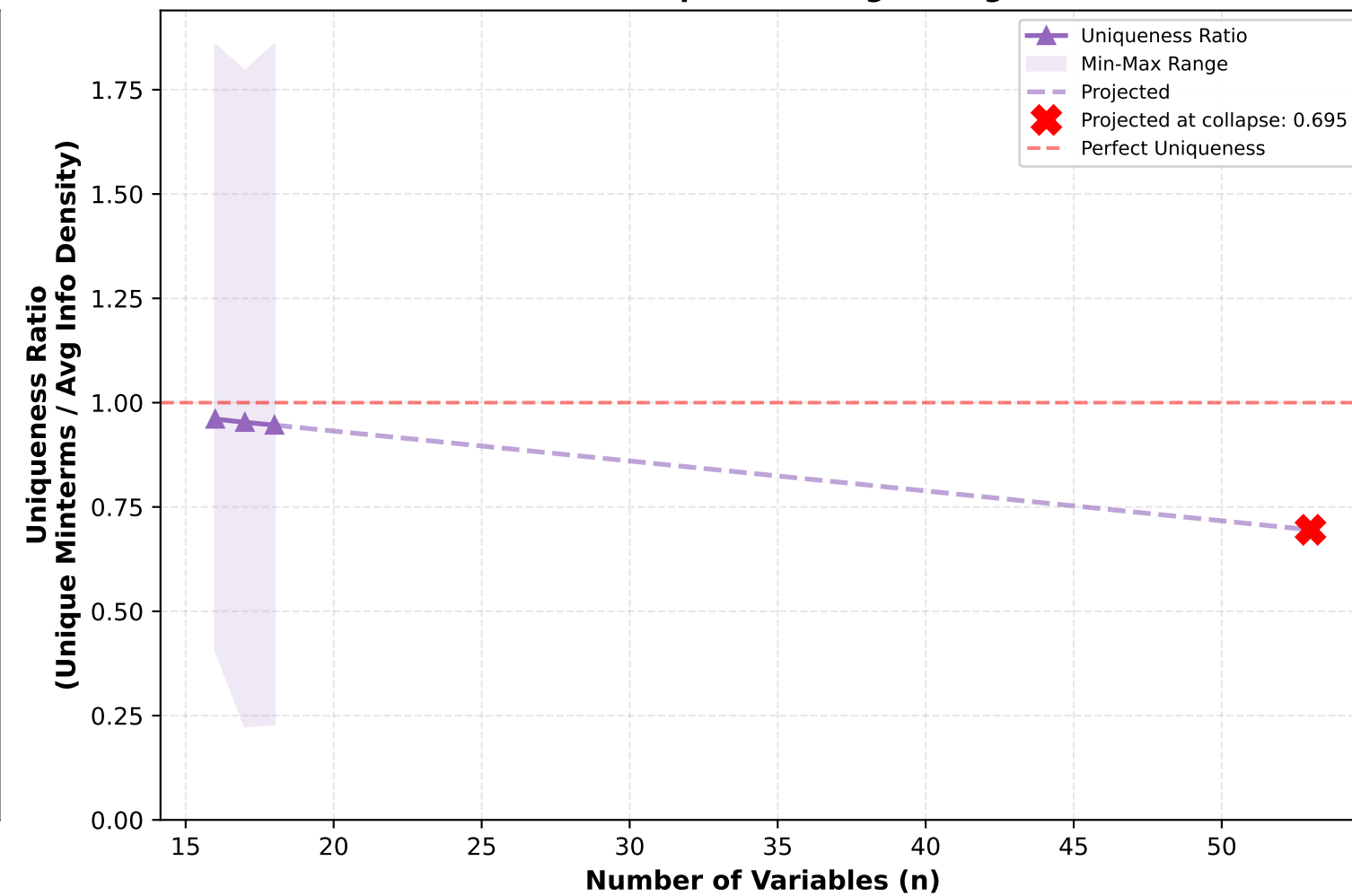
Information Density (avg $\pm$ range)



Cluster Growth

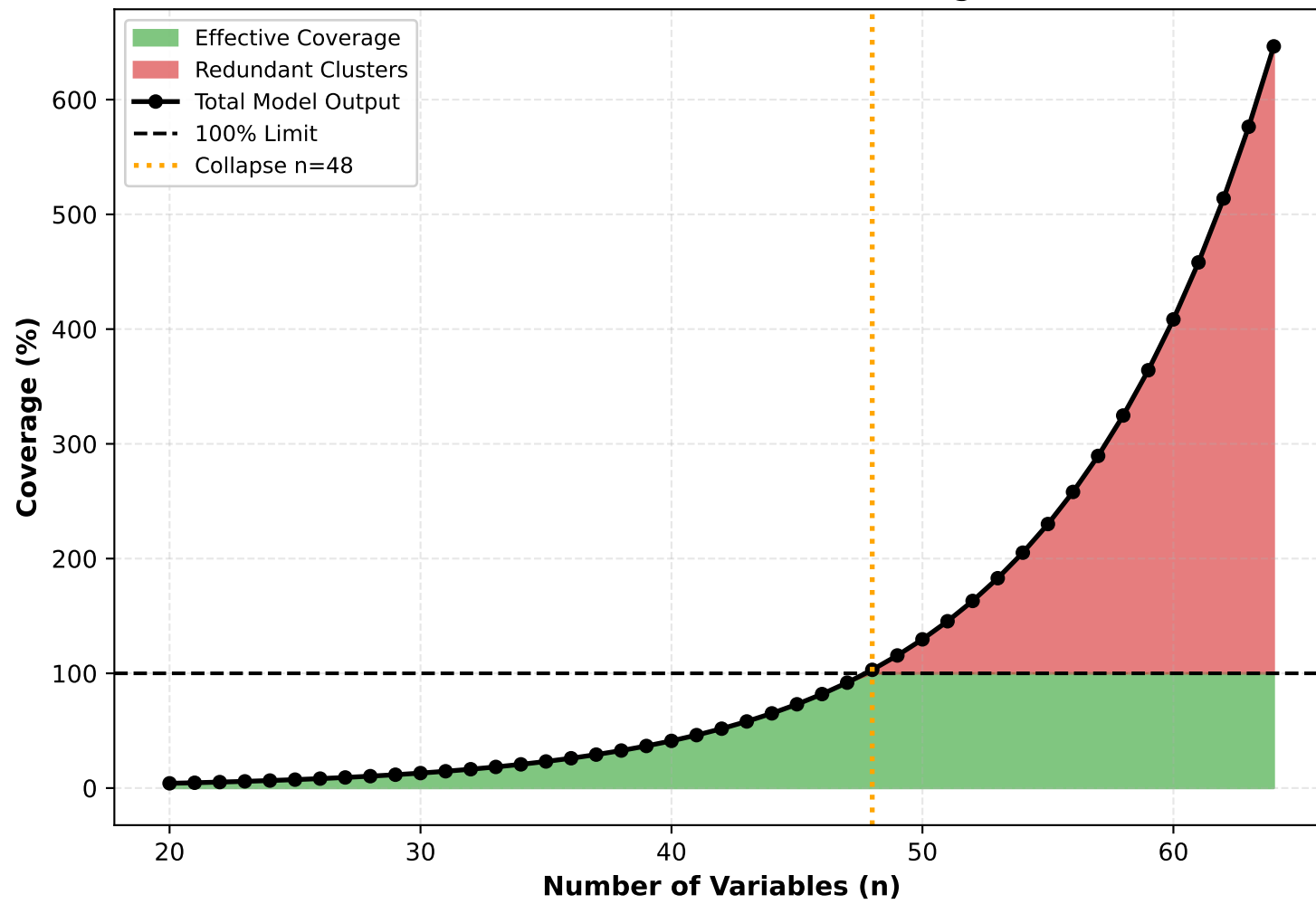


Cluster Uniqueness (avg $\pm$ range)

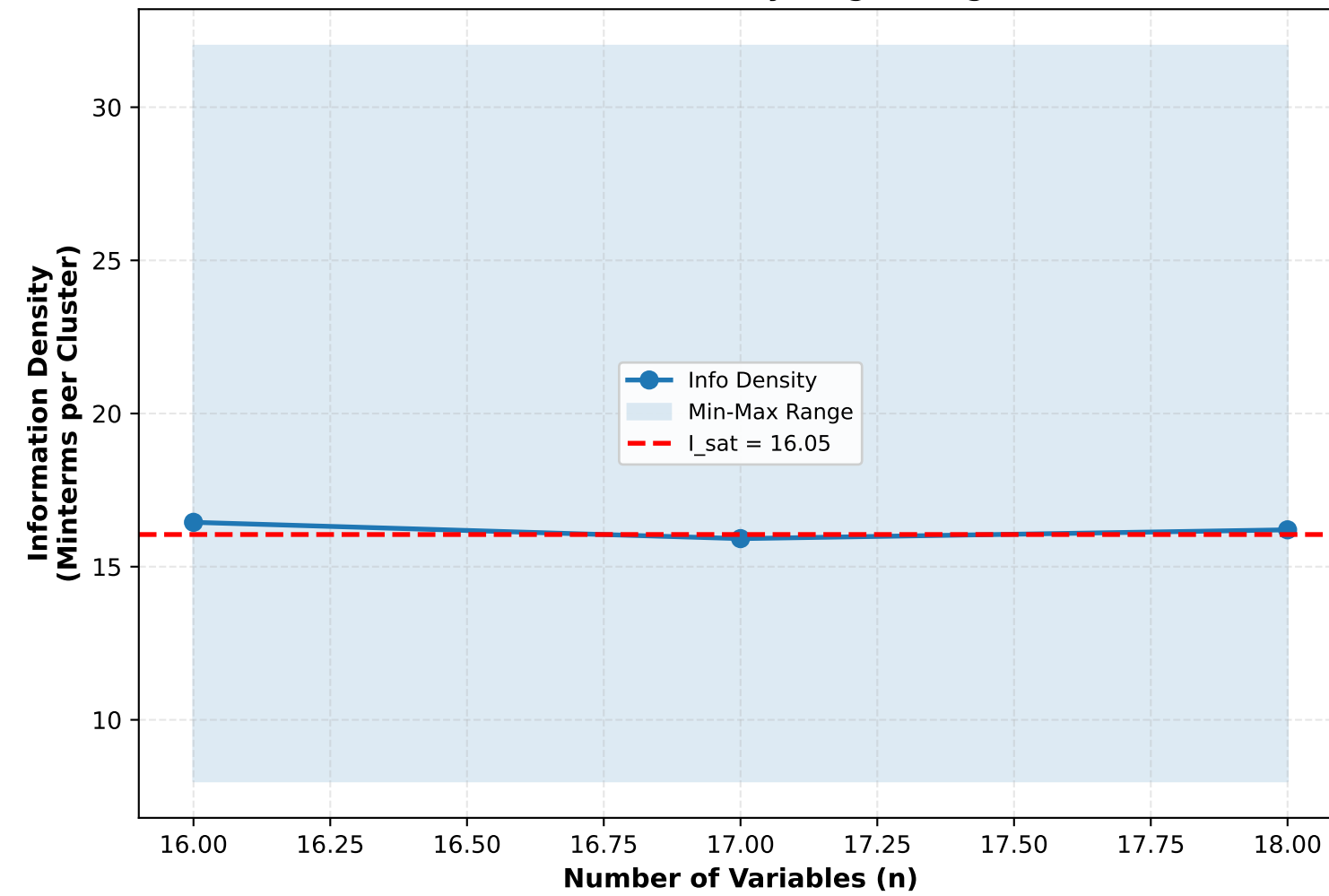


4D Cluster Analysis - Density 0.7

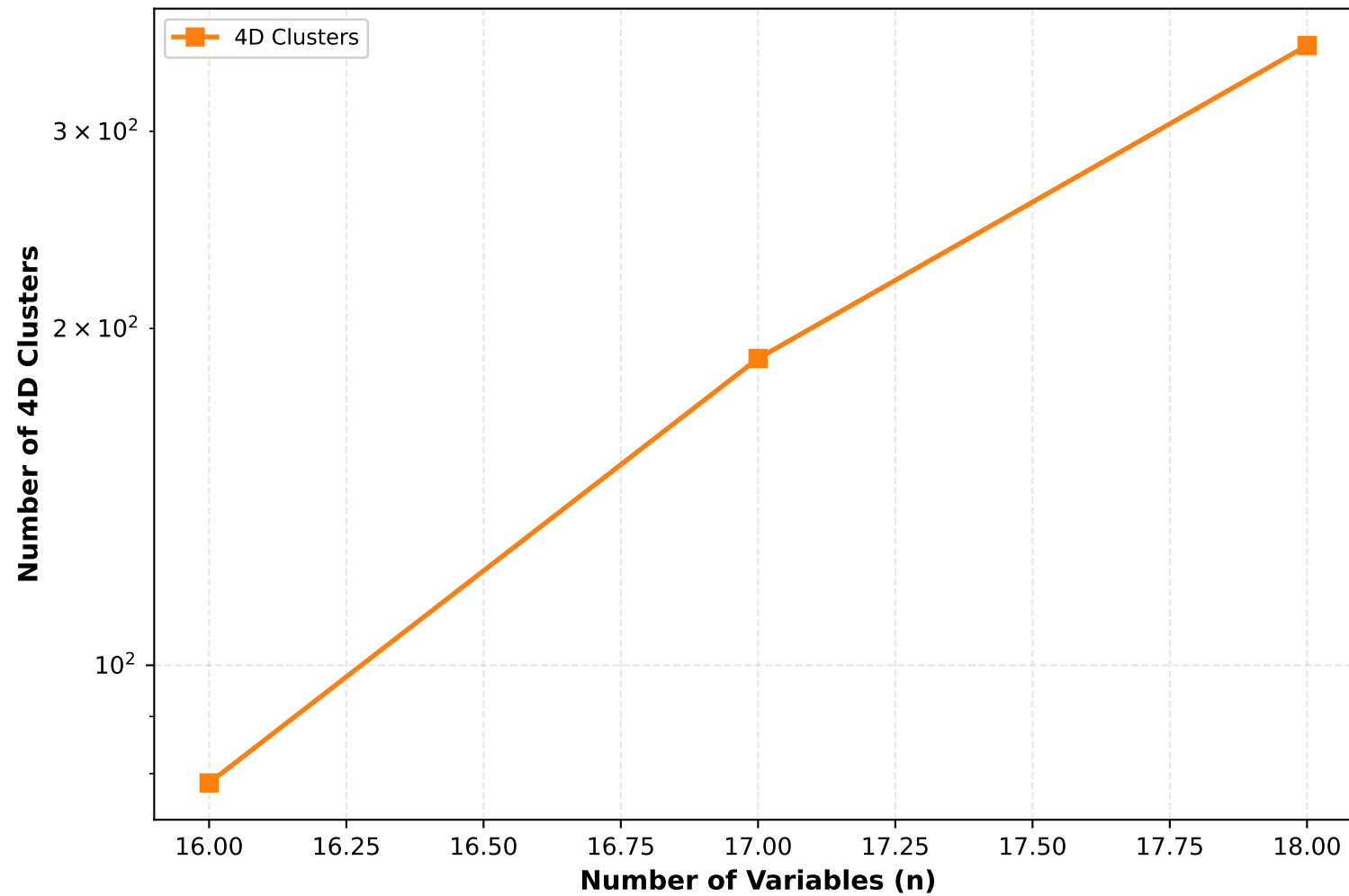
Effective vs Redundant Coverage



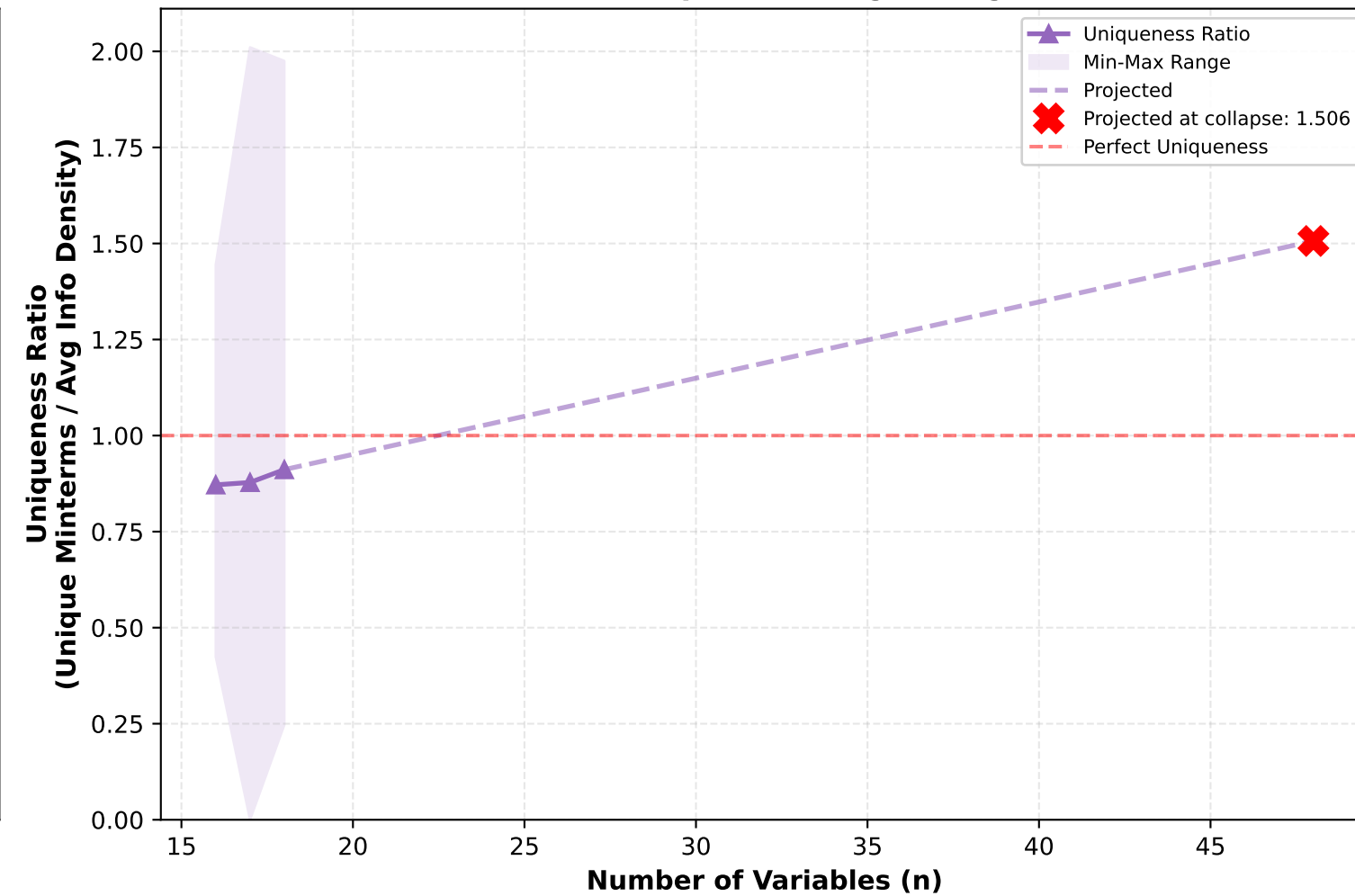
Information Density (avg $\pm$ range)



Cluster Growth

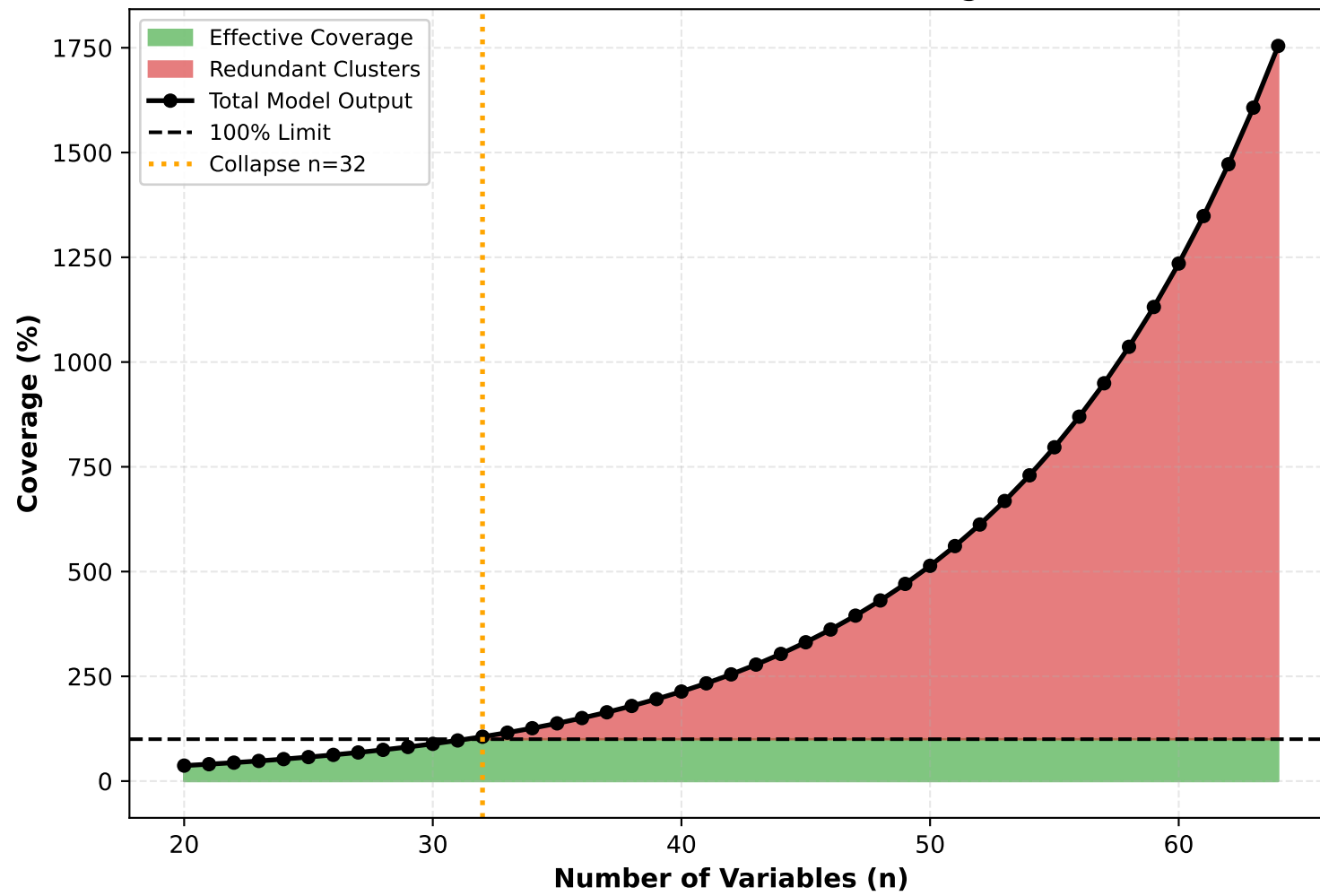


Cluster Uniqueness (avg $\pm$ range)

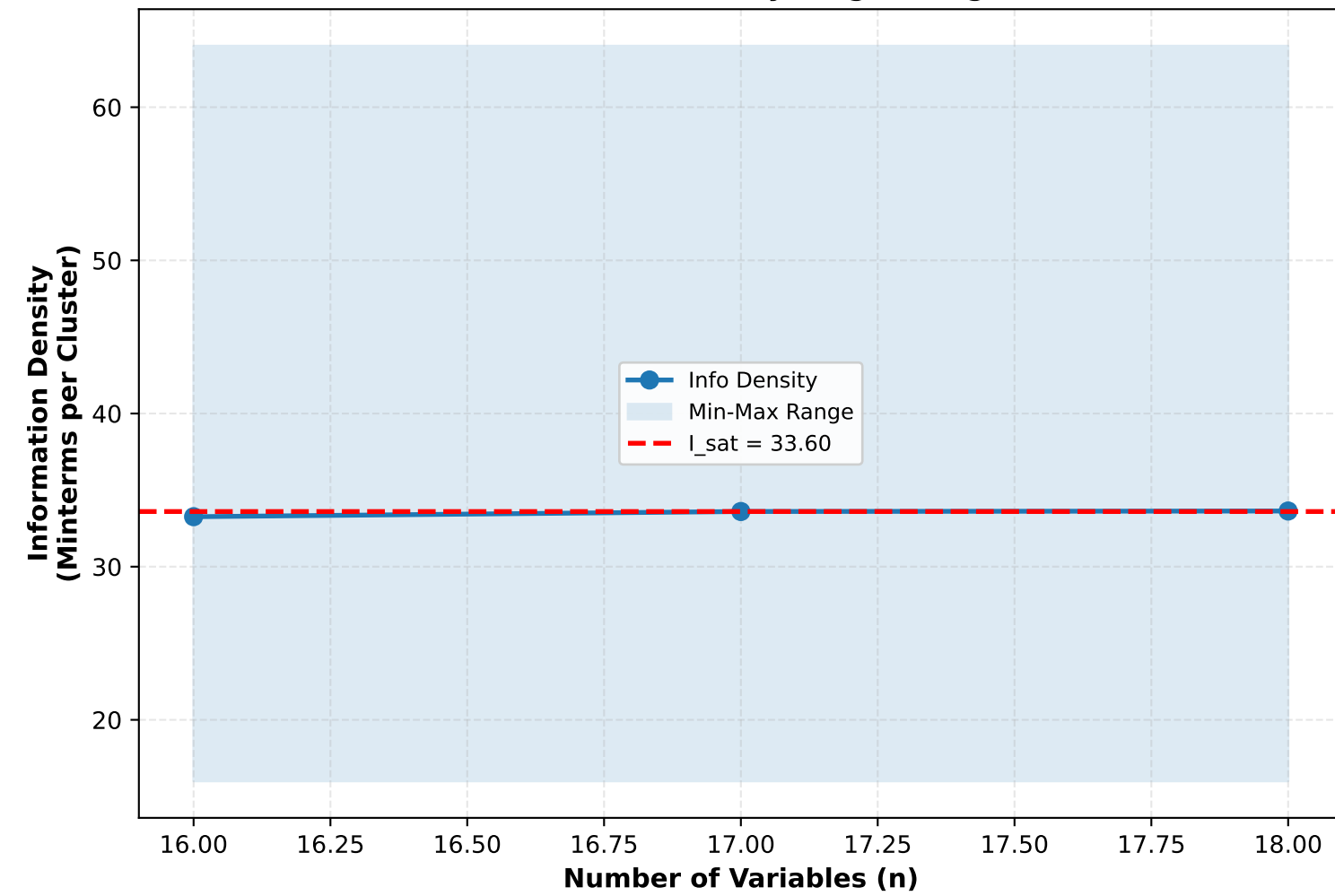


4D Cluster Analysis - Density 0.9

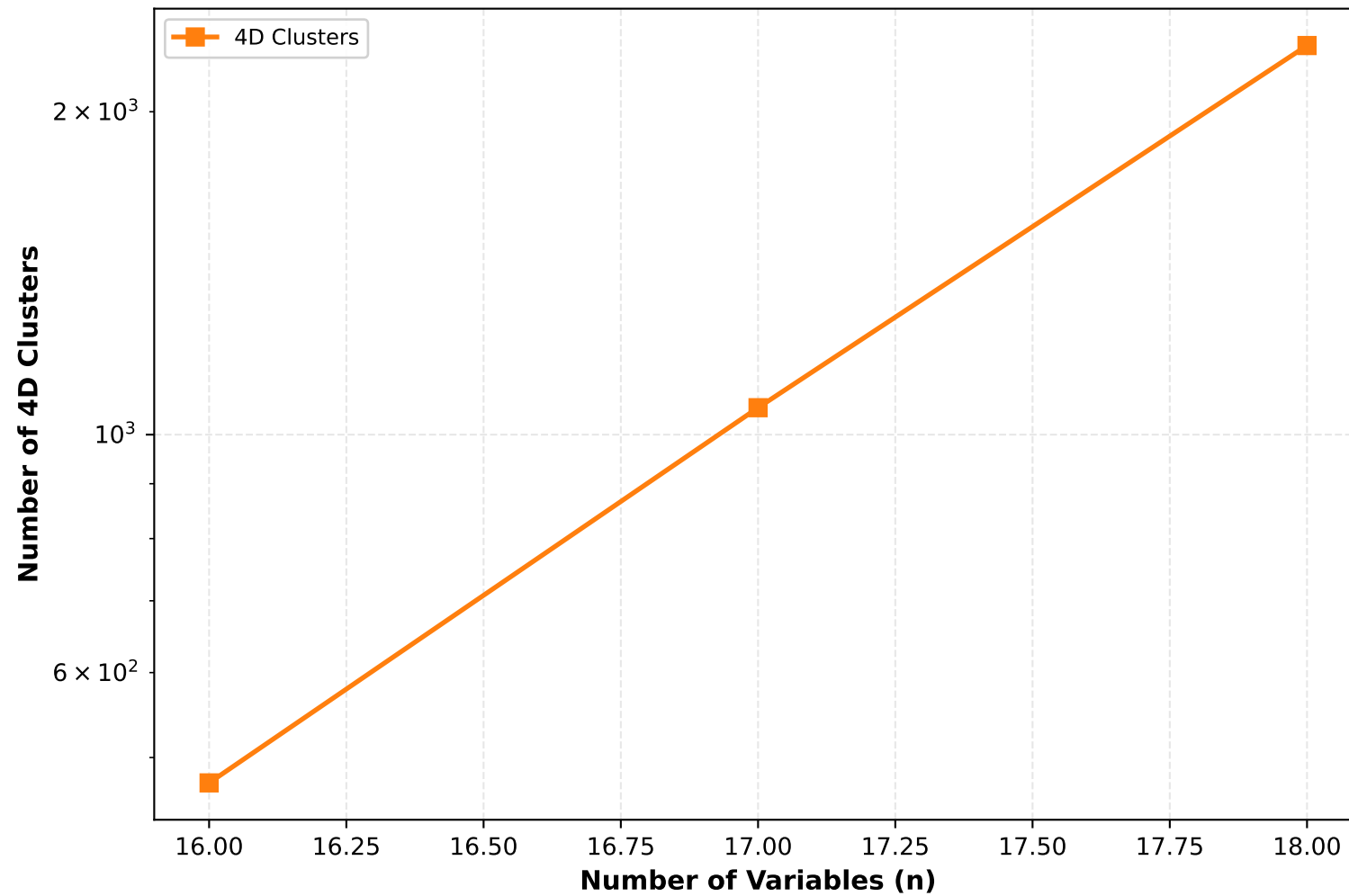
Effective vs Redundant Coverage



Information Density (avg $\pm$ range)



Cluster Growth



Cluster Uniqueness (avg $\pm$ range)

